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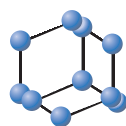
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REVIEW ARTICLE

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Review on *Stenotrophomonas maltophilia*: An Emerging Multi-drug-resistant Opportunistic Pathogen

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Abstract: *Stenotrophomonas maltophilia* is an opportunistic pathogen that results in nosocomial infections in immunocompromised individuals. These bacteria colonize on the surface of medical devices and therapeutic equipment like urinary catheters, endoscopes, and ventilators, causing respiratory and urinary tract infections. The low outer membrane permeability of multidrug-resistance efflux systems and the two chromosomally encoded β -lactamases present in *S. maltophilia* are challenging for arsenal control. The cell-associated and extracellular virulence factors in *S. maltophilia* are involved in colonization and biofilm formation on the host surfaces. The spread of antibiotic-resistant genes in the pathogenic *S. maltophilia* attributes to bacterial resistance against a wide range of antibiotics, including penicillin, quinolones, and carbapenems. So far, tetracycline derivatives, fluoroquinolones, and trimethoprim-sulfamethoxazole (TMP-SMX) are considered promising antibiotics against *S. maltophilia*. Due to the adaptive nature of the intrinsically resistant mechanism towards the number of antibiotics and its ability to acquire new resistance via mutation and horizontal gene transfer, it is quite tricky for medicinal contribution against *S. maltophilia*. The current review summarizes the literary data on pathogenicity, quorum sensing, biofilm formation, virulence factors, and antibiotic resistance of *S. maltophilia*.

Keywords: Antibiotic resistance, biofilm, virulence factors, *Stenotrophomonas maltophilia*, antimicrobial, β -lactams.

1. INTRODUCTION

Stenotrophomonas maltophilia, formerly known as *Pseudomonas maltophilia*, was grouped in the *Xanthomonas* genus in 1983 and placed under *Stenotrophomonas* in 1993 [1]. It is a Gram-negative, non-fermentative, aerobic bacteria [2-6]. *S. maltophilia* is motile, and its motility depends on polar flagella. It is an environmental bacterium found in soil, air, water, microflora, and rhizospheres [7-10]. It is a catalase-positive, oxidase-

negative, and lysine decarboxylase bacteria [11]. It displays positive reactions to extracellular DNase. *S. maltophilia* commonly causes acute respiratory tract infections (RTI), soft tissue infections, necrotizing otitis, keratitis, meningitis, arthritis, bacteremia, endocarditis, peritonitis, ophthalmic diseases, and osteochondritis [12-17]. Patients on dialysis are more susceptible to the pathogen as they are immunocompromised and have increased exposure to artificial accesses like prosthetic grafts or peritoneal catheters; this can act as a source of *S. maltophilia*. The colonization of the bacteria in the fluids used in hospitals, irrigation solutions, and invasive hospital devices facilitate the average host defense to cause infection, which acts as a vehicle for *S. maltophilia*. *S. maltophilia* has de-

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veloped intrinsic resistance to various antibiotics using mutation and horizontal gene transfer, including β -lactams, quinolones, aminoglycosides, tetracycline, and disinfectants. Over the years, *S. maltophilia* has emerged to be a multi-drug resistant pathogen. The genome of *S. maltophilia* encodes for multidrug-efflux pumps, β -lactamases, and aminoglycoside-modifying enzymes [18]. Multidrug-efflux pumps and low permeability of the outer membrane are significant determinants of the intrinsic antibiotic resistance of *S. maltophilia* [10]. *S. maltophilia* can also acquire resistance by taking resistance genes located on integrons, transposons, and plasmids [13]. SmeABC is involved in acquired, but not intrinsic, resistance to β -lactams, aminoglycosides, and quinolones. Overexpression of SmeDEF, SmeVWX, and SmQnr are the main mechanisms responsible for fluoroquinolone resistance in *S. maltophilia* [19]. Resistance to β -lactams arises from the expression of two inducible β -lactamases: L1 and L2. L1 is an Ambler class B Zn^{2+} -dependent metalloenzyme that hydrolyses all categories of β -lactams except the monobactams. L2 is an Ambler class A serine active site β -lactamase (an ESBL) inhibited by clavulanic acid [20]. These two chromosomal β -lactamases are usually, but not always, induced when cells are exposed to β -lactams. The mechanisms employed by *S. maltophilia* that mediate resistance to aminoglycosides primarily involve aminoglycoside-modifying enzymes and multidrug efflux pumps. These enzymes include the aminoglycoside acetyltransferase AAC (6')-Iz and the aminoglycoside phosphotransferase APH (3')-IIc [21]. Both of which confer low-level resistance to aminoglycosides, except gentamicin. Mutations in the quinolone-resistance-determining region of genes encoding topoisomerases (*gyrA*, *gyrB*, *parC*, and *parE*) are associated with the effective mechanism of quinolone resistance employed by bacteria [22]. The resistance of Gram-negative bacteria to sulfonamides is mainly conferred by the acquisition of either *sul1* or *sul2*, encoding dihydropteroate synthases. The primary phenicol resistance determinant in *S. maltophilia* is *floR*, which encodes an exporter protein of the MFS family that mediates resistance to chloramphenicol and florfenicol [23].

S. maltophilia is a common source of antibiotic-resistant infections acquired in hospitals, with a 37.5 [24] percent attributable mortality rate in immunocompromised patients, which makes it

unique. *S. maltophilia* infection is more likely to occur in immunocompromised patients, as well as those with cancer or pre-existing inflammatory lung disorders like cystic fibrosis [25]. In addition, illnesses acquired in the community have been documented. Resistance to several antimicrobial classes, including beta-lactam antibiotics, aminoglycosides, cephalosporins, and macrolides, makes the treatment options limited through the non-heritable adaptive mechanisms, genetic material acquired through horizontal transfer, and intrinsic resistome. Even in COVID-19 patients, *S. maltophilia* was also seen as a secondary infection, increasing mortality risk [26]. Thus, it is evident that *S. maltophilia* is an emerging drug-resistant pathogen under the radar and needs special approaches for treatment alongside antimicrobial therapy.

2. EPIDEMIOLOGY

S. maltophilia is an environmental bacterium; its presence in infection is more common worldwide [27]. In several infectious diseases, *S. maltophilia* acts as an etiological agent [28, 29]. The trace of clinical manifestation reveals that bacteria's ability to change its virulence factors and traits makes the pathogen more successful. The virulence factor for colonization and morbidity is possessed by both environment and clinical strains [30]. World health organizations reported that developing resistance mechanisms made antibiotics increasingly ineffective as drug resistance spreads globally, leading to more difficulty in treating common infections. The attributed mortality rates of *Stenotrophomonas maltophilia* infections in pneumonia range between 25% to 75%, 26% to 28% in case of nosocomial bloodstream infections, and crude mortality rates of *S. maltophilia* bacteremia range from 21% to 69% [31, 32]. The mortality rate will increase sharply until adequate treatment and prevention of drug-resistant infections. The overall findings of the study state that the *S. maltophilia* strains majorly fall under human and environmental sources. The strains of Sm6 lineages represent the primary human pathogens. They are found to be the human invasive, noninvasive, or respiratory strains, likely to have enhanced virulence and adaptations to human infections. Previous genomic study reports unveiled that *S. maltophilia* strains are incredibly diverse at the nucleotide levels, which causes a challenge for population-wide analyses. In Germany, the occurrence of *S. maltophilia* as a noso-

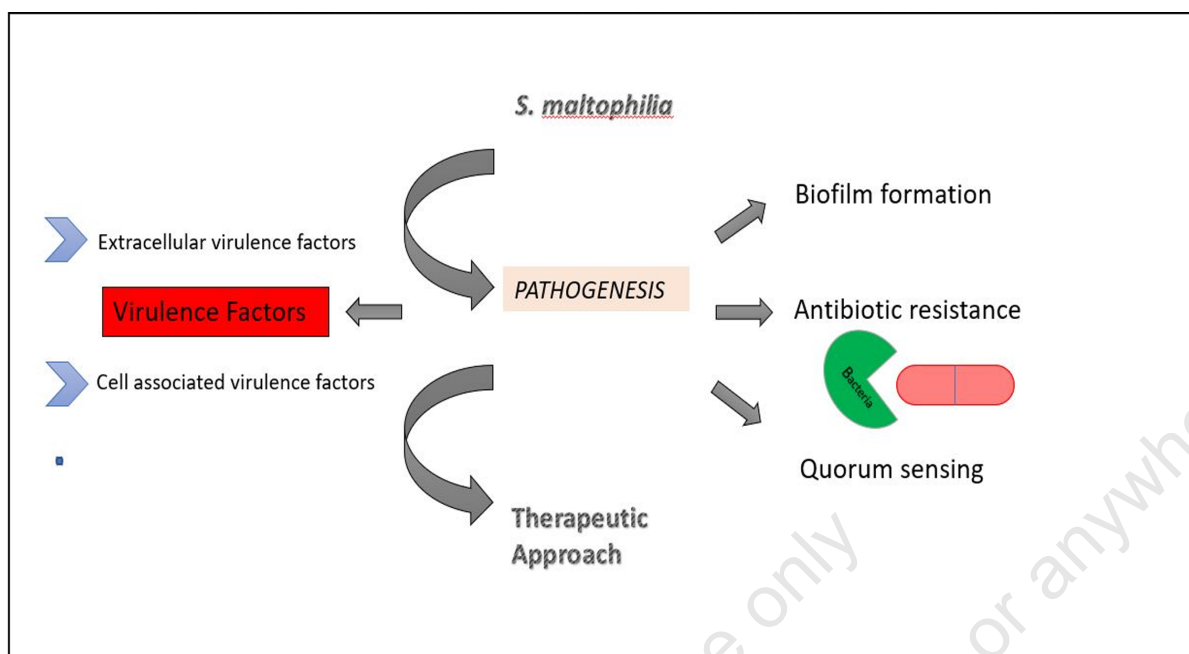


Fig. (1). Illustrative model depicting factors contributing to the pathogenesis of *S. maltophilia* which can be tapped to unravel several therapeutic approaches in the treatment of infections caused by this particular bacteria. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

comial infection in Intensive Care Unit isolates is as high as 165 isolates per 1000 patients between 2001 and 2004 in some study locations. A surveillance study of 6 weeks in Washington, USA, reported a 9.4% frequency from stool samples [33]. In Los Angeles, the USA, an outbreak of *S. maltophilia* blood colonization among allogeneic bone marrow transplant patients is observed [34]. A study conducted in Turkey from June, 2000, to December, 2001, shows that 44 strains isolated as etiological agents were from 41 hospitalized patients. In a study, 12 strains of the bacteria caused three outbreaks in the studied area [35]. *S. maltophilia* is known to cause pneumonia, urinary tract infection (UTI), septic shock, ophthalmologic infection, Surgical Site Infection (SSI), and colonization of medical implants among immunocompetent patients [36]. The bacteria act as an etiological agent of otitis externa and pyomyositis in immunocompromised patients [37, 38]. *S. maltophilia* is reported to be found in about 3.7% of patients discharged from the hospital and is the third most common non-fermenting Gram-negative bacilli responsible for hospital-acquired infections [39]. *P. aeruginosa* and *Acinetobacter* spp. are the first and second most common bacilli causing human health-related infections, respectively. A case of non-healing wounds in an immunocompetent patient due to the colonization of the *S. maltophilia*

was reported in India [40]. In Taiwan, an outbreak of drug-resistant meningitis resulting from *S. maltophilia* infection was reported [41]. In France, a multidrug-resistant strain that was pacemaker-infective was reported to cause endocarditis, and in Saudi Arabia, cases with complications resulting in otitis externa were reported. Some of these infections associated with *S. maltophilia* have been fatal [42-44]. An 11-year study conducted until 2008 among pneumonia patients revealed a recovery rate of 3.1% in *S. maltophilia* infections [45]. The recovery rate in Southern America was the least, about 2.3%, whereas the patients from the EU had a recovery rate of about 3.2%. The highest recovery percentage was noted in the patients from the United States (3.3%) [45]. A study was conducted for the epidemiological analysis of the clinical and molecular characteristics of *S. maltophilia* infection in a Chinese teaching hospital [46]. Multilocus Sequence Typing (MLST) analysis stated that 61 different sequence types (STs) were found (including 45 novels STs), and MLST did not show significantly dominant STs. Pulsed-Field Gel Electrophoresis (PFGE) results indicated that 93 isolates could be divided into 73 clusters, and they also showed weak genetic linkages between isolates. The resistance rates to trimethoprim/sulfamethoxazole (TMP/SMX) and levofloxacin were 9.7 and 4.3%, respectively, and all isolates

were susceptible to minocycline. Four virulence gene loci *StmPr1*, *StmPr2*, *Smf-1*, and *Smlt3773* were positive in 79.6, 91.4, 94.6, and 52.7% of the isolates, respectively. Three biofilm genes *rmlA*, *spgM*, and *rpjF* were positive in 82.8, 92.5, and 64.5% of the isolates. Another study aimed to characterize the molecular epidemiology of *S. maltophilia* strains associated with an outbreak in the Children's Hospital of México Federico Gómez (HIMFG). Typing using pulsed-field gel electrophoresis (PFGE) showed a 52% genetic diversity with seven pulsotypes designated P1-P7 among all *S. maltophilia* strains. Three pulsotypes (P2, P3, and P7) were identified among the clinical and environmental *S. maltophilia* strains and associated with two types (STs), namely, ST304 and ST24. Moreover, 80% (24/30) of the strains exhibited resistance mainly to tetracycline, 76.66% (23/30) to trimethoprim-sulfamethoxazole, and 23.33% (7/30) to the extended-spectrum β -lactamase (ESBL) phenotype. Furthermore, the *pilU*, *hlyIII*, and *rmlA* genes were identified in 96.6% (29/30), 90% (27/30), and 83.33% (25/30) of the samples, respectively. Additionally, 76.66% (23/30) of the *S. maltophilia* strains exhibited high swimming motility, 46.66% (14/30) showed moderate biofilm formation capacity, 43.33% (13/30) displayed moderate twitching motility, and 20% (6/30) exhibited high adherence [47]. Another study investigated the epidemiology of *S. maltophilia* emergence in the CF Centre of Roscoff, Western France, a region of the high prevalence of CF in Europe. *S. maltophilia* was isolated in 90 patients (42 males, 48 females). The mean age at first colonization was 24.4 ± 13.5 years. Annual prevalence since 2013 was high (16-17.9%) but stable. This high prevalence is mainly due to a high rate of intermittent colonization. Only 2.8% of CF patients showed chronic colonization, with significantly more frequent co-colonization by methicillin-susceptible *Staphylococcus aureus* ($P < 0.0001$) and *Pseudomonas aeruginosa* ($P < 0.05$). During chronic colonization, *S. maltophilia* acquired resistance to cotrimoxazole and β -lactams [48].

3. PATHOGENESIS

In some cases, *S. maltophilia* acts as a true pathogen [49, 50]. *S. maltophilia* is an aetiological agent for pyomyositis in an immunocompromised individual [51]. The *S. maltophilia* infection predominantly occurs in hospitalized immunocompe-

tent adults, especially patients in intensive care units (ICU) who are more vulnerable. *S. maltophilia* is featured by colonization, followed by tissue invasion [52, 53]. Prolonged hospitalization leads to the invasion of bacteria into the host. The malignancy of the infection depends on the duration of hospitalization. Before the *S. maltophilia* invasion, the length of hospitalization is generally between 11.5 days and 24 days [54, 55]. Contamination in medical devices acts as a vector for bacterial infection. The prosthetic materials like plastics and metals present in the transplants facilitate the infection. Intravenous drug abuse and exposure to a wide range of antibiotics are also known to cause diseases [56]. Colonization of *S. maltophilia* occurs on the tubes' surfaces used in mechanical ventilation, urinary catheters, endoscopes, and suction catheters. The motility of *S. maltophilia* depends on the polar flagella. The most commonly available polar flagella are unflagellate, biflagellate, and multi-flagellate, which have a prominent role in forming a biofilm. The colonization of *S. maltophilia* happens through the biofilm, facilitating the attachment of bacteria to the abiotic environment. Genes *rmlA*, *rmlC*, and *xanB* coding for biofilm production are significant for colonization [57]. Quorum sensing is always accompanied by biofilm formation. The QS molecule detects the number of cells present in the host [58] and responds accordingly through gene regulation. *S. maltophilia* K279a possesses a Diffusible Signal Factor (DSF) dependent system. *rpjF* which contributes to the synthesis of DSF, and *rpjF* is present in *rpf* operon, facilitating the regulation of virulence factors [59]. QS regulation includes the N-Acyl homoserine lactones pathway and DSF-QS pathway. This pathway helps in the control of biofilm formation [60], bacterial motility [61], and virulence [61]. Several enzymes, such as deoxyribonuclease, protease, ribonuclease, gelatinase, and lipase, play an essential role in pathogenesis [62, 63]. *StmPr1* encodes for the enzyme protease in *S. maltophilia*. The protease breaks down the protein components like collagen, fibronectin, and fibrinogen, resulting in local tissue damage and bleeding [64]. Cyclic diguanosine monophosphate (c-di-GMP) regulator is involved in contributing to bacterial pathogenicity. The *BsmR* acts as a negative regulator by degrading c-di-GMP in *S. maltophilia*. *BsmR* results in 349 flagella genes [63], making the *S. maltophilia* cells move apart and preventing

biofilm formation. Some other essential macromolecules are obtained through specific pathways regulated by small RNAs expression. These regulatory genes are the most challenging factor for antibacterial agents (Fig. 1).

3.1. Biofilm Formation

Biofilm is the agglomerate of microorganisms in which cells that are frequently embedded within a self-produced matrix of Extracellular Polymeric Substances (EPSs) adhere to each other and surface [65]. Biofilms form on the abiotic surfaces and tissues; there can be high, moderate, and low levels of biofilm formation. The extracellular matrix of polysaccharides and proteins embeds a group of bacterial cells attached to the surface to form a biofilm [66]. Different types of biofilms include periphytons, stromatolites, flocs, *etc.* [67, 68]. Biofilm production of *S. maltophilia* was first reported in 1996. Authors demonstrated that unspecific adhesion to negatively charged surfaces, such as glass and Teflon, was favoured by *S. maltophilia* compared to *Pseudomonas putida*. The extracellular matrix with 65% biofilm formation is associated with hospital-acquired infections and contamination of medical devices and equipment, leading to *S. maltophilia* [69]. Briefly, planktonic bacteria first adhere to a biotic or abiotic surface; this first step often requires the presence of flagella or pili. Once they become firmly attached, the cells migrate together using a process of twitching motility or recruit planktonic cells and grow to form microcolonies. Sessile bacteria excrete extracellular polymeric substances (EPSs), creating a heterogeneous matrix interspersed with open water channels [70]. Since biofilm can also form on moist surfaces and is resistant to water-soluble disinfectants, respiratory tubes, intravenous lines, catheters, nebulizers, dialysis equipment, dental equipment, and hospital plumbing systems are often colonized. Interestingly, *S. maltophilia* adheres along intercellular junctions of human epithelial respiratory cells on CF-derived epithelial monolayer culture and mucin, highlighting the significance of *S. maltophilia* pathogenesis in the airways of CF patients [70].

Numerous genes are associated with biofilm formation. The *spgM* gene encodes a bifunctional enzyme with both phosphoglucosyltransferase and phosphomannosyltransferase activities that are involved in

LPS biosynthesis, playing an important role in biofilm formation [71]. The *rmlA* gene encodes glucose-1-phosphate thymidyltransferase involved in the LPS/EPS-coupled biosynthetic pathway. The *rpfF* gene, encoding the DSF (diffusible signal factor) synthase, regulates virulence, such as motility, extracellular proteases, LPS, and biofilm production [71]. *BsmR* (a response regulator gene) degrades c-di-GMP, responsible for biofilm formation and swimming motility of the bacteria. It is also involved in the representation of potential targets for new antibacterial agent synthesis [71].

The formation of biofilm contributes to the multidrug resistance mechanism. It also helps in the resistance against the host immune system. Colonization of *S. maltophilia* on the host surfaces can be prevented by controlling the virulence factors of the bacteria. Bacterial biofilm structure and composition result in inherent resistance to antimicrobial agents [72]. The biofilm matrix acts as a barrier to antimicrobial agents. Some antibiotics (*e. g.*, ciprofloxacin, tobramycin, or β -lactams) are effective against the outer layer of the biofilm, which is metabolically active. Colistin kills the inner layer of the cells of biofilm with low metabolic activity. The exopolysaccharides of the matrix are negatively charged and can bind to positively charged antibiotics trying to reach cells in the biofilm. Uncharged antibiotics, such as β -lactams, are unlikely to attach to exopolysaccharides.

Moreover, some proteins are expressed preferentially in biofilm cells [73, 74]. In biofilm formation, the production of proteins related to antimicrobial resistance and virulence increases. Several mechanisms are responsible for resistance, including difficulty in penetration of the antimicrobial agent through the biofilm matrix, the altered growth rate of biofilm organisms, and nutrient limitations [75]. Bostanghadiri *et al.* showed biofilm formation significantly correlated with ceftazidime and SMX/TMP resistance. Similarly, biofilm formation has been shown to be associated with resistance to different antibiotics, including ceftazidime, piperacillin/tazobactam, cefepime, ticacillin/clavulanic acid, aztreonam, and gentamicin. In this study, they investigated the relationship between biofilm formation and the presence of its related genes *rpfF*, *spgM*, and *rmlA*. Sixty-three isolates of *S. maltophilia* strains had all genes

studied, while only 81 strains carried the *spgM*. Overall, the results revealed that the presence of the *spgM* significantly promoted biofilm formation [76].

3.2. Antibiotic Resistance

Several mechanisms contribute to multidrug resistance in *S. maltophilia*. Low outer membrane permeability, antibiotic modifying enzymes, and multidrug efflux pumps are involved in the antibiotic resistance of *S. maltophilia*. It shows resistance to a wide range of antibiotics, such as β -lactam antibiotics, tetracyclines, TMP-SMX, chloramphenicol, fluoroquinolones, macrolides, aminoglycosides, cephalosporins, carbapenems, and polymyxins [77-79].

Structural biology plays a major part in the resistance of *S. maltophilia* against a wide range of antibiotics. This includes the tripartite membrane efflux pumps, whose structure influences the rate of drug efflux across the membrane of bacteria. For example, SmeT has two extensions at the N and C termini relevant to its function. Interestingly, SmeT possesses the smallest binding pocket. *In*

vitro studies revealed that SmeT binds to a 28-bp pseudopalindromic region, forming two complexes that overlapped the promoters of SmeT and SmeDEF [80].

Other experimental approaches which can be useful to have a better understanding of the multidrug resistance nature of *S. maltophilia* would be the omics approach to unfold different players involved and the treatment that can be undertaken to subdue the resistance.

3.2.1. Trimethoprim-Sulfamethoxazole Resistance

Trimethoprim-Sulfamethoxazole is the treatment of choice due to its activity *in vitro* and good clinical outcome in treating *S. maltophilia* infections with a susceptibility rate higher than 90% [81, 82] (Fig. 2). However, TMP-SMX resistance has been reported in Australia, Europe, the USA [83], Argentina [84], Taiwan [85], and Malaysia [86]. A study has revealed that out of 58 patients analyzed, 11 patients (18.9%) required a therapy alternative to TMP-SMX, indicating that almost one-fifth of the patients were infected with TMP-

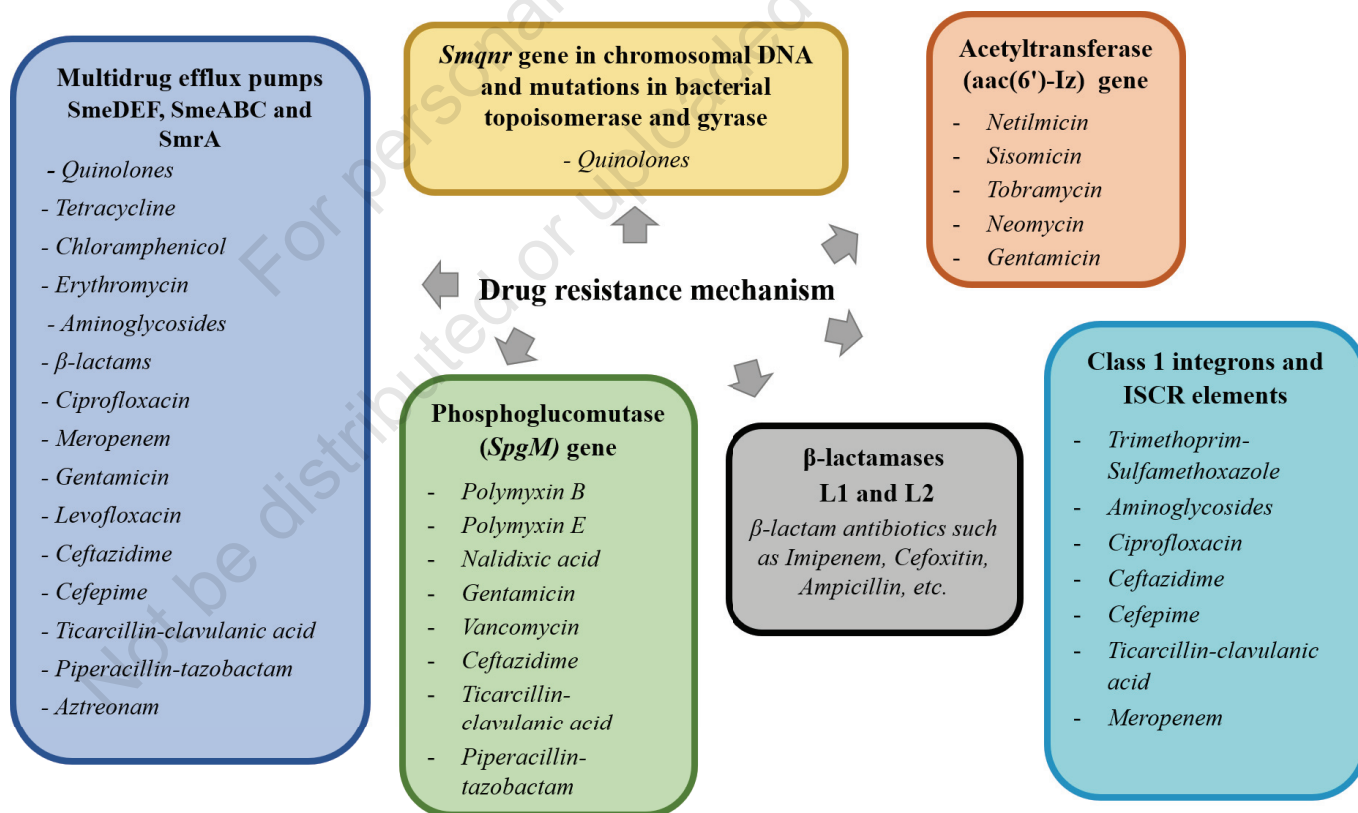


Fig. (2). Multi-drug resistance mechanism in *Stenotrophomonas maltophilia*. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

SMX resistant *S. maltophilia* [87]. Overexpression of the efflux pump SmeDEF shows that the susceptibility of TMP-SMX is reduced, whereas its presence contributes to the intrinsic resistance to TMP-SMX [87]. Integrons are non-self-mobilizable elements with integrase enzymes for inserting antibiotic gene resistance cassettes between the highly conserved nucleotide sequences. These gene cassettes are integrated and excised by the site-specific recombinase IntI1 encoded by class 1 integrons, recognizing integron-associated attI1 site and 59- base element family of sites in the gene cassettes [87]. It plays a vital role in the multidrug resistance of *S. maltophilia* by facilitating horizontal gene transfer for acquiring a variety of resistance genes from the surrounding environment and assembling them in the form of tandem gene cassettes, thereby conferring antimicrobial activity resistance to multiple drugs. Plasmids and transposons contribute to the dissemination of integrons as it lacks enzymes for transposition [88]. TMP-SMX resistance has been attributed to two alleles of the *sul* genes (*sul1* and *sul2*). The integrase gene (*intI1*) is present at the 5'-conserved end, and the *sul1* gene, which encodes for resistance to sulfonamides, is present in the 3'-conserved end of the class 1 integron [89]. In most cases, the *sul2* gene has been found on plasmid DNA and can also be part of chromosomal DNA in *S. maltophilia* TMP-SMX-resistant isolates. The *sul2* gene is closely linked to insertion sequence common region (ISCR) elements in *S. maltophilia* [90]. The *dfrA* gene cassettes encoded for the dihydrofolate reductase enzyme are also located in the class 1 integrons and contribute to TMP-SMX-resistance [91]. The effect of mobilization of the *sul* genes due to class 1 integrons and ISCR elements impacts the TMP/SMX therapy. The *sul* genes, in combination with the *dfrA* gene, contribute to TMP-SMX resistance [91].

3.2.2. Aminoglycoside Resistance

Aminoglycosides are potent natural or semisynthetic antibiotics derived from Actinomycetes. They act through the inhibition of protein synthesis. Different classes of aminoglycosides include streptomycin, gentamycin, tobramycin, neomycin, and amikacin. Resistance to aminoglycosides has evolved over the years, which involves mechanisms underlying enzymatic drug modification and efflux mediated resistance [88]. The resistance

mechanism in *Stenotrophomonas maltophilia* primarily encompasses the modification of aminoglycosides (apart from gentamycin) by aminoglycoside acetyltransferase (AAC (6')-Iz) that confers resistance to netilmicin, amikacin, sisomicin, tobramycin [92], and aminoglycoside phosphotransferase APH (3')-Iic, thereby conferring resistance to paromomycin, butirosin, kanamycin, and neomycin [93]. Chromosomally carried *aph(3')-Iic* gene from *Stenotrophomonas maltophilia* clinical isolate K279a encodes an aminoglycoside phosphotransferase enzyme that confers intrinsic resistance to aminoglycosides [93]. AAC (6')-Iak is another 6'-N- aminoglycoside acetyltransferase identified later in a multi-drug resistant strain of *S. maltophilia* that mitigates its susceptibility against aminoglycosides, such as sisomicin, tobramycin, arbekacin neomycin, and kanamycin [94]. The PhoPQ sensor-regulator two components system regulates the outer membrane structure and confers virulence and antibiotic-resistant *S. maltophilia*. Susceptibility to aminoglycosides increases when *phoP* is inactivated, as it reduces the expression of the SmeZ efflux pump and contributes to alterations in the membrane permeability [95]. Furthermore, the most inducible resistance mechanism is the SmeYZ efflux system, where SmeZ is an RND-type efflux pump. It has been shown that mutations in Ribosomal Protein RplA or treatment with ribosomal acting antibiotics activate aminoglycoside Efflux Pump SmeYZ in *S. maltophilia* [96]. These SmeABC, SmeOP-TolCsm, and MacABCsm are associated with resistance [97]. Interestingly, *clpA* and *htpX* are found to be the primary determinants responsible for intrinsic aminoglycoside (AG) resistance. It has been observed that the inactivation of *clpA* and *htpX* compromised protease-mediated intrinsic aminoglycoside resistance. It weakened SmeYZ pump-mediated aminoglycoside resistance, signifying HtpX and ClpA as potential AG adjuvant targets for treating *S. maltophilia* infections [98]. Temperature is another factor that regulates aminoglycoside resistance in *S. maltophilia*. Due to alterations in the outer membrane fluidity at a temperature of 37°C, these bacterial species are known to be more susceptible to aminoglycosides [99]. At a temperature of 30°C, the LPS in the outer membrane has a more prominent O-antigen component, which inhibits interactions with antibiotics. This makes the bacteria more resistant at 30°C than at 37°C [99].

3.2.3. β – lactam Resistance

S. maltophilia has two inducible β -lactamases that are encoded chromosomally. β -lactamase L1 is a homotetramer of 118 kDa and belongs to an Ambler class B zinc-dependent metalloenzyme [100]. β -lactamase L2 is a clavulanic acid-sensitive cephalosporinase that belongs to an Ambler class A serine active site β -lactamases [101]. AmpR is a transcriptional regulator in the L2 upstream region, as part of an *ampR*-L2 module, and controls β -lactamases expression at the transcriptional level [101]. The binding of AmpR influences the expression of lactamase to the intergenic sequence placed between the L2 and the *ampR* gene [102]. It is required for both basal and induced expression of the L1 gene. For the L2 gene, AmpR acts as a weak repressor in the absence of β lactam antibiotics and as an activator in the presence of β lactam antibiotics. AmpC is a transcriptional regulator linked to peptidoglycan recycling and is associated with multiple regulatory genes, such as *ampG*, *ampD*, and *ampR*. AmpC expression is repressed when AmpR is bound to UDP-*N*-acetyl muramic acid-pentapeptide and activated when AmpR is bound to anhydro-*N*-acetylmuramyl-peptide [103]. Other proteins associated with the cell wall include AmpD and AmpG. AmpD is a protein involved in cleaving the cell wall components into 1, 6-anhydro muramic acid and peptide. In a study, two *ampD* homologs: *ampDI* and *ampDII*, were identified in *S. maltophilia*. *ampDI* is an anhydro-*N*-acetylmuramyl-*l*-alanine amidase responsible for both L1 and L2 β -lactamases regulation, whereas *ampDII* is not responsible for the regulation of the β -lactamases [103]. AmpG is involved in recycling additional proteins present in the cell wall and is also responsible for transporting degraded components of cell walls into the cytoplasm. Unlike other bacterial species, an *ampN*-*ampG* operon is required to express lactamases L1 and L2 in *S. maltophilia* and *ampR*, *ampN*, *ampD*, and *ampC* genes [104]. New studies reveal that in iron-depleted conditions, *S. maltophilia* exhibits elevated β -lactamase activity. The expression of the *ampR* gene increases when the β -lactam challenge occurs, leading to the binding of AmpR to 1,6-anhydroMurNAc-peptides and acting as an activator to induce β -lactamase expression [105]. AmpI encodes an inner membrane protein involved in iron export. Inactivation of *AmpI* inhibits iron ex-

port and leads to an elevated intracellular iron level. This inhibition results in reduced siderophore synthesis and a strengthened iron storage system. When *S. maltophilia* is subjected to β -lactam antibiotics, the L1 and L2-*ampI* operon are upregulated, increasing ROS levels. The L1 and L2 β -lactamases protect bacterial cells by hydrolyzing β -lactams. This environment induces AmpI to function as a cytosolic iron exporter, limiting ROS toxicity levels in the bacteria. Thus, AmpI contributes to mitigating ROS stress caused by β -lactam in *S. maltophilia* [106]. The inconsistent occurrence of inducible β -lactamases inhibits β -lactams and is responsible for high resistance to this type of antibiotic.

3.2.4. Quinolone Resistance

Quinolone resistance primarily arises from two mechanisms, such as efflux pumps and a chromosomally encoded *qnr* gene (*Smqnr*) that protects both gyrase and topoisomerase IV [107]. Mutations in the quinolone-resistance-determining region of genes encoding topoisomerases (*gyrA*, *gyrB*, *parC*, and *parE*) are associated with the effective mechanism of quinolone resistance in bacteria. Moreover, *gyrA* mutation in *S. maltophilia* does not result in fluoroquinolone resistance. High-level resistance to fluoroquinolones, including some of the most active fluoroquinolones developed, such as ciprofloxacin, can be attained without topoisomerase QRDR mutations [108]. In *S. maltophilia*, mutations in topoisomerases encoding genes are not associated with quinolone resistance [108]. Furthermore, *S. maltophilia* harbors a novel quinolone resistance gene, *Smqnr*, encoded by the chromosome rather than the plasmid-mediated *qnr* gene [109]. Plasmid-mediated quinolone resistance (PMQR) is a newly identified antimicrobial resistance mechanism. It is mainly mediated by antibiotic resistance *qnr* gene, including subtypes of *qnrA*, *qnrB*, *qnrC*, *qnrD*, and *qnrS*, in which *qnrA* is the most common subtype [110]. Overexpression of *smeDEF* and *smeVWX* genes are critical in resistance development and can lead to high-level fluoroquinolone resistance [111]. In 2018, it was reported that MsfA is a novel fluoroquinolone efflux pump of *S. maltophilia*, and plasmid-mediated overexpression of *mfsA* in wild type *S. maltophilia* K279a increased resistance not only to paraquat but also to second-generation

fluoroquinolone antibiotics, *i.e.*, ciprofloxacin, norfloxacin, levofloxacin, and ofloxacin [112].

3.2.5. Multidrug Efflux Pumps

A significant factor contributing to multidrug resistance is the capacity of the bacteria to export drugs through the intrinsic activity of the efflux pumps [113]. The genome of *S. maltophilia* encodes multidrug efflux pumps that can be classified as eight predicted resistance nodulation cell division (RND)-type efflux systems with SmeABC, SmeDEF, SmeVWX, SmeIJK, SmeYZ, and SmeOP-TolCSm characterized [114-119] and SmeMN and SmeGH uncharacterized [120], ATP-binding cassette (ABC)-type (SmrA, FuaABC, and MacABCsm), and Qsuperfamily (MFS)-type (EmrCABsm, MfsA). Overexpression of these efflux pumps is associated with reduced antibiotic susceptibility [121]. SmeDEF, SmeIJK, SmeOP, SmeYZ, and ABC efflux pump MacABCsm have been shown to play a role in intrinsic resistance to different compounds [121]. The overexpression of SmeABC provides resistance to β -lactams, aminoglycosides, and quinolones, although only the deletion of the OMP of the system, SmeC, affects the susceptibility to several antimicrobial drugs. SmrA, the first ABC-type efflux pump identified in *S. maltophilia*, is responsible for acquired resistance to fluoroquinolones, tetracycline, doxorubicin, and multiple dyes. FuaABC, a fusaric acid (5-butylicpicolinic acid, a mycotoxin) efflux pump, a member of a subfamily of the ABC-type family, is induced by fusaric acid, and its overexpression leads to fusaric acid resistance. The MacABCsm efflux pump allows intrinsic resistance to aminoglycosides, macrolides, and polymyxins and contributes to oxidative and envelope stress tolerance as well as biofilm formation. The MFS-type pump EmrCABsm is involved in extruding hydrophobic compounds, including the antibiotics nalidixic acid and erythromycin [122]. A novel MFS efflux pump (MfsA) plays an essential role in mediating resistance to paraquat [123], as well as to antibiotics, such as aminoglycosides (kanamycin, streptomycin, and neomycin), cephalosporins (cefazolin and cefalexin), fluoroquinolones (ciprofloxacin, norfloxacin, levofloxacin, and ofloxacin), the macrolide erythromycin, rifampicin, tetracycline, and chloramphenicol [123]. SmeABC is involved in acquired resistance to β -lactams, aminoglycosides, and quinolones. SmeDEF is involved in intrinsic

and acquired (in the condition of overexpression) resistance to quinolones, tetracyclines, macrolides, chloramphenicol, novobiocin, and SXT, as well as acquired resistance to triclosan [124]. SmeVWX mediates acquired resistance to chloramphenicol, quinolones, and tetracyclines and, when overexpressed, increases susceptibility to aminoglycosides [125]. SmeYZ mediates intrinsic resistance to aminoglycosides and SXT [126, 127], while SmeIJK is involved in reduced intrinsic susceptibility to gentamicin, amikacin, tetracycline, minocycline, ciprofloxacin, and leucomycin [128,129]. SmeIJK also mediates acquired resistance to levofloxacin when overexpressed alone or in coordinate hyperproduction with SmeYZ [129]. SmeOP-TolCSm efflux pump is associated with decreased susceptibility to nalidixic acid, doxycycline, aminoglycosides (amikacin and gentamicin), and macrolides (erythromycin and leucomycin).

3.2.6. Tetracycline Associated Drugs

Tetracycline resistance is the consequence of active efflux of the antibiotic, and it is associated with resistance to quinolones and chloramphenicol, but not to aminoglycosides or beta-lactam antibiotics. SmrA and SmeDEF is a complex formed by an efflux pump located on the inner membrane (SmeE), an outer membrane protein (SmeF), and a periplasmic membrane fusion protein (SmeD). It is involved in resistance to tetracyclines [130]. A serendipitous mutation in the regulatory smeT gene, such as A/T mutation at position 498, could lead to overexpression of smeDEF and multiple resistances to tigecycline, doxycycline, and levofloxacin [130]. Recently, a study on clinical isolates of *S. maltophilia* in Beijing showed that smeDEF efflux pumps gene was the common genetic determinant of resistance to tigecycline, doxycycline, and levofloxacin [131].

3.3. Quorum Sensing

S. maltophilia cells communicate through a regulatory mechanism that detects, produces, and responds to the extracellular signal molecules known as autoinducers. The bacterial cell density increases and makes the autoinducers accumulate in the environment. Due to this increase in concentration, the cells sense a change in their population, regulating target genes [132]. In *S. maltophilia*, the Quorum system functions mainly through Diffusible Signal Factor (DSF) signaling, Ax21

system, and Acyl Homoserine Lactone (AHL) signaling systems. The DSF signaling functions through genes located in the *rpf* (regulation of pathogenicity factors) cluster. [132] The genes that include enzymes for DSF synthesis, recognition, and signal transduction are located in two adjacent operons and are transcribed convergently. One of the operon codes for *RpfF* synthase and *RpfB* FA ligase and the other operon codes for *RpfC* sensor kinase and *RpfG* cytoplasmic regulator functions as a two-component system. In *S. maltophilia*, the variation in *rpf* cluster variant strains, *rpf-1* and *rpf-2*, plays an essential role in DSF signaling. The *rpfF* gene has two alleles, *rpfF-1* and *rpfF-2*. While in *rpf-2* strain, the DSF production requires either an extra copy of *rpfF-2* or the absence of *RpfC-2*, which is the sensor/repressor component, *rpf1* strain expresses DSF production in standard growth conditions. In *S. maltophilia* strain K279a, which has *rpfF-1* allele, deletion of *rpfF* gene results in decreased protease secretion, lower resistance to antibiotics, and a change in lipopolysaccharide [133]. In the wild-type WR-C, *rpfF* or *rpfB* gene deletion decreases ferric citrate receptor (FecA) expression [133]. Conversely, in *S. maltophilia* strain M30, an *rpfF-2* strain, a mutation in the *rpfF* gene does not affect biofilm formation, virulence, or the motility of the bacteria [134]. Nevertheless, it has also been observed that when *rpf-1* and *rpf-2* strains co-inhabit, the virulence and DSF production enhance in the whole population [134]. It can be understood from this mechanism that, in the presence of DSF-producing bacterial strains, *rpf-2* strain triggers DSF signaling [134]. A recent study has revealed that based on the *rpf* variant, resistance to certain antibiotics is influenced. It has been reported in this study that the *rpf-1* subpopulation displays higher levels of resistance to the β -lactam antibiotics, ceftazidime and ticarcillin, and strains of the *rpf-2* group showed higher resistance levels to colistin. This effect is attributed to the ability to produce DSF by different strains [135]. Ax21 is an outer membrane protein extensively conserved in plant pathogenic *Xanthomonas*-associated species, including *S. maltophilia* [136]. A study on Smlt0387 and Smlt0184 from the *S. maltophilia* K279a variant, homologous to Ax21 of Xoo strain PXO99A, found that the deletion of smlt0387 led to reduced motility, biofilm formation, and reduced tolerance to tobramycin, an aminoglycoside antibiotic. A

general practice to test the association of a signaling protein to a cell-cell signaling system is that if the phenotypic effect caused by deleting the signal synthetase can be reversed by adding the signal molecule exogenously, the association can be confirmed [137]. Concerning Ax21, the addition of synthetic Ax21 protein restored motility in smlt0387 deleted cells [138]. A correlation study on Ax21 homolog secretion and DSF proved that DSF induced the production of outer membrane vesicles. Ax21 homologs associated with OMV also tend to increase [139]. This possibly denotes that the Ax21 homolog is dependent on DSF signaling. However, the exact role of Ax21 is not appropriately studied in *S. maltophilia* and requires more studies to conclude the relevance. N-acyl homoserine lactone (AHL) is another quorum system by which *S. maltophilia* cells communicate [140-142]. AHL signals are associated with LuxI-type synthases and sensed by LuxR-type transcriptional regulators [143, 144]. The genome of *S. maltophilia* strain K279 encodes 15 putative LuxR solos. Only SmoR protein encoded by the *smlt1839* gene presents the typical N-terminal AHL-binding domain C-terminal helix-turn-helix (HTH) DNA-binding domain [145]. *In vitro* AHL-binding assays confirmed that SmoR from strain E77 binds to AHL signal oxo-C8-HSL, regulating swarming motility. The *S. maltophilia* E77 parental strain strongly stimulates swarming motility in the presence of a *P. aeruginosa* supernatant that contains a high level of AHLs, including oxo-C8-HSL, a synthetic AHL, indicating that SmoR senses AHL signals produced by neighboring bacteria [145]. It is noted that *S. maltophilia* is closely related to the environment and can communicate effectively with other species. Antibiotics that specifically target the quorum system could mitigate the virulence effects of this pathogen.

3.4. Virulence Factors

The factors or molecules synthesized by bacterial cells that facilitate host organisms' damage through colonization, adhesion, and invasion are known as virulence factors. Two types of virulence factors are extracellular and cell-associated virulence factors. The extracellular virulence factor includes esterases, lipases, phospholipases, nucleases, hyaluronidase, heparinase, siderophores, cytotoxins, and hemolysins. In contrast, cell-associated virulence factors include lipopolysac-

charides, pili, fimbriae, non-pilus adhesins, and flagella.

3.4.1. Cell Associated Virulence Factors

3.4.1.1. Lipopolysaccharide

As *S. maltophilia* is a Gram-negative bacterium, it possesses a low permeable outer membrane composed of lipopolysaccharides. LPS is responsible for colonization and provides resistance to complement-mediated cell killing. The LPS is composed of lipid-A, core oligosaccharides, and O-specific polysaccharides. Lipid-A consists of phosphorylated glucosamine residues with N-fatty acyl and O-fatty acyl components [146]. Core oligosaccharide consists of D-Glucose, D-Galactose, D-Mannose, D-Galacturonic acid, D-Galactosamine, 3-Deoxyoctulosonic Acid, and L-Glycero- α -D-Mannoheptose as the main components [147]. The elements of O-antigen include rhamnose, fucose, xylose, and glucose [147]. The two operons, *rmlBACD* and *xanAB*, produce LPS, *rmlC*, and *rmlA* to synthesize O-antigen and *xanB* biosynthesis in the core region of LPS O-antigen [147]. The *spgM* gene is involved in the biosynthesis of enzymes responsible for both phosphoglucomutase and phosphomannomutase activities. These enzymes play an essential role in the assembly of O-polysaccharide chains.

Incomplete LPS can alter the production of biofilm. *S. maltophilia*, with insufficient LPS length, results in reduced virulence in the rat lung model of infection. The mutant strain (K2049) of *S. maltophilia* with *spgM* chromosomal knockout prevents colonization in rat lungs. The mutation or inactivation of the *spgM* gene results in a lower yield of high-molecular-weight O-antigen, unlike the parental strains. The study on mutant strains inoculated in the rat lung tissues reveals the importance of LPS as a virulence factor in bacteria. The alterations may change bacterial cells' susceptibility to antimicrobial agents (cationic peptides and aminoglycosides) in LPS. Temperature change alters the chemical composition of LPS, which, in turn, limits the exposure. *spgM* chromosomal knockout mutant strain (K2048, K2049) of *S. maltophilia* exhibited sensitivity to certain antibiotics like polymyxin B, polymyxin E, nalidixic acid, gentamicin, and vancomycin, unlike their parental strains. Mass spectrometry reveals a high degree of heterogeneity among the lipid A compo-

nents from different bacteria's clinical isolates. The virulence of *S. maltophilia* depends on the variations between the strains [147].

3.4.1.2. Pili

Pili is also known as fimbriae. The colonization of *S. maltophilia* was initiated by adherence to epithelial cells with the help of adhesins. Bacterial adhesins are of two types: pili (fimbriae) or non-pilus (afimbrial adhesins) [148]. Different pathogenic bacteria possess a variety of bacterial adhesins. Fimbriae help in the binding of *S. maltophilia* to host cells or abiotic surfaces and maintain biofilm stability [149]. Fimbriae form bridges between bacterial cells and host cells or abiotic surfaces, resulting in colonization. High-resolution scanning electron microscopy demonstrates bridges between the bacterial cells and the target host cells or abiotic cells. Unlike the parental strain, the mutant strain with defective production of fimbriae is less virulent. This is due to the ineffective binding of bacteria to the target host cells [149-150].

The *Smf-1* gene is involved in hemagglutination, biofilm formation, and cell adhesion. The presence of the *Smf-1* gene in clinical isolates of cystic fibrosis patients has been revealed in recent studies. The strains of environmental origin do not have a trace of the *Smf-1* gene, and it supports the fact that the *Smf-1* gene is involved in respiratory tract colonization in CF patients [151]. The anti-*Smf-1* antibodies inhibited the agglutination of animal erythrocytes, biofilm formation by *S. maltophilia*, and adherence to HEp-2 cells.

Apart from the *Smf-1* fimbrial operon, two other pili/fimbrial clusters in the genome of *S. maltophilia* are identified: the one responsible for biofilm formation and adherence TadE-like cluster [152], and the other has significant similarity with the giant cable pilus locus of *Burkholderia cenocepacia*. The genes encoding type four pilus machinery are scattered throughout the *S. maltophilia* genome [153]. Type four pili (TFP) has various functions like twitching motility, adherence to host cells, and modulation of the target cells [154, 155]. There is no significant correlation between the TFP-mediated motility, biofilm formation, and adhesion. All these structures can act as biofilm adhesins qualitatively and independently of their function [156]. TFP is not responsible for the ini-

tial adherence of *S. maltophilia* to respiratory tract cells of CF patients but plays an essential role in forming a biofilm. A specific mechanism TFP can control biofilm formation in CF lungs.

3.4.1.3. Flagella

Flagella in *S. maltophilia* are used for motility and adhesion. Flagella are highly immunogenic structures that adhere to the host cells by interconnecting the bacterial cells and the surface, visualized by high-resolution electron microscopy of bacterial monolayers on synthetic surfaces [157]. Electron microscopy reveals the variation in the number of flagellar structures in *S. maltophilia*. Flagellin, a globular protein, is the major structural subunit of flagella [157].

Flagella binding of *S. maltophilia* to the mucous membrane prevents bacterial cells from moving in the respiratory tract, which affects the initiation of the infection. The bacteria's ability to adhere to synthetic surfaces impacts the initiation of opportunistic infections in hospitalized patients and immunocompromised individuals. The protection of bacteria from the host immune system is due to the biofilm formed on the mucous membrane through flagella-mediated attachment [158].

The direct adhesion of clinical *S. maltophilia* strains to the mouse tracheal mucus aided by flagella is revealed in a study [158]. Adherence with defective flagella is compared with that of the wild-type parental strains to examine its ability to adhere to IB3-1 cells (human bronchial epithelial cells from CF patients) by flagella. The result of the study concludes that the loss of flagella leads to reduced bacterial adhesiveness. It does not lead to a complete lack of adhesiveness in bacteria. The same survey showed that the motility of bacterial study is not correlated with adhesion to IB3-1 cells. The study's combined results suggest that flagella's role in adherence is not based on its functionality. Adhesion is also carried out by other structures.

The amount of biofilm formed on the polyvinyl chloride surfaces of a mutant strain of *S. maltophilia* with defective flagella and its wild-type parental strain is compared with the adherence in a study; the results show no substantial differences. These findings indicate that flagella do not affect biofilm formation but its significance in early adherence stages [159]. Another study compared the

isolates from CF patients with the isolates from non-CF patients. The study stated that the flagella-mediated motility plays a role in biofilm formation only in isolates of *S. maltophilia* from CF patients. Still, it is not required for its initiation. From the studies, the author concludes that *S. maltophilia* has a distinctive phenotype known as the 'CF phenotype' and is selected by adapting to the extreme environment of the lung in CF [159].

3.2.2. Extracellular Virulence Factors

3.4.2.1. Proteases

Bacterial cells use proteases to cleave extracellular sources to obtain amino acids according to requirements. These protein degrading enzymes have been identified as host virulence factors [160]. *S. maltophilia* secretes extracellular serine proteases against nematodes [161]. It has been observed that different serine proteases are involved in nosocomial and nematocidal infections. This suggests that the nosocomial strains encode enzymes from genes present in the non-hospital environment. Proteolytic enzymes are involved in nutrient acquisition, invasion, and destruction of host tissue, pathogen colonization, and host immune system modulation [162]. Among the clinical isolates of *S. maltophilia*, protease activity varies depending on variations in substrates and media used for its propagation and the exoenzyme potential of specific infectious strains [163]. In immunocompromised patients, the proteolytic activity of *S. maltophilia* can cause fulminant hemorrhagic pneumonia and can degrade connective tissue, especially fibronectin and collagen. Histological examination disclosed bleeding in fatal infection cases due to damage to the lung epithelium [164, 165]. Other reports indicated that *S. maltophilia* is involved in massive hemorrhagic conditions in the small intestine and the subclavian artery region, accompanied by severe damage to the tissue [166, 167]. In cystic fibrosis patients infected with *S. maltophilia*, the release of proteolytic enzymes contributes to detrimental lung tissue effects.

StmPr1 encoded by the *StmPr1* gene is the major extracellular protease responsible for degrading human serum proteins and connective tissues under *in vitro* conditions. It inactivates the components of the host immune defense. *StmPr1* mainly affects the tight junctions, adherens junctions, and ECM connections. *StmPr2* gene is the structural

gene that encodes for the minor extracellular protease *StmPr2*. Mutations in this gene result in the absence of protease activity. There are variations in the nucleotide sequences of *StmPr1* genes, but the *StmPr2* gene is highly conserved. Two allelic variants of *StmPr1* have been identified, whereas the *StmPr2* locus has been detected in all strains. *StmPr1* and *StmPr2* are products of the Xps type 2 secretion system, one of the six secretion systems found in Gram-negative bacteria. This system causes actin rearrangement, cell detachment, cell rounding, and cell death in *S. maltophilia*. Both enzymes are responsible for the caseinolytic, serine-protease, and gelatinase activity [168]. In addition to these two enzymes, studies suggested that additional proteins contribute to the Xps-mediated degradation of interleukin-8 and extracellular matrix, including collagen, fibrinogen, and fibronectin. *StmPr3* is another Xps secreted-serine protease that acts along with *StmPr1* and *StmPr2*. Another enzyme, *StmPr4*, is also identified in small quantities and is used in commercial detergent preparation [169].

Enzyme gelatinase produced by *S. maltophilia* is one of the essential biochemical features of its species. In other gram-negative bacteria, this enzyme participates in biofilm formation, degradation of host tissue, and inactivation of the complement system to avoid immune response [170]. The role of gelatinase in *S. maltophilia* is relatively not known. A case of leukemia patients with ecthyma gangrenosum due to bacteremia by *S. maltophilia* has been reported [171]. The exoenzyme profile of the isolate includes both gelatinase and elastase. The elastase is responsible for the elastic lamina degradation of blood vessels, enhancing bacterial cells' release in subcutaneous tissue [171]. Fibrinolysin is another proteolytic enzyme that is mainly found in Gram-positive bacteria and *S. maltophilia*. It acts as a spreading factor, but data are insufficient for *S. maltophilia*. According to studies published earlier, fibrinolysin degrades the fibrin clots formed around skin lesions and facilitates the diffusion of *S. maltophilia* into the host [172, 173].

3.4.2.2. Heparinase

The enzyme heparinase is responsible for the degradation of the host's extracellular matrix, present in the respiratory system, and contributes to pathogenicity [174]. Heparinase has a substrate

named Heparin sulfate, which is the extracellular proteoglycan component of bronchial infections. Enzyme activity is found more among sputum and tracheal aspirates. The heparinase enzyme initiates the pathological processes, which leads to tracheo-bronchial disease [175]. Flores-Díaz *et al.* identified variability in the heparinase enzyme activity according to the isolation source with the other enzymes in the extracellular matrix [174]. Comparatively, more heparinase-producing strains were found in medical devices [174].

3.4.2.3. Hyaluronidase

Hyaluronidases break down hyaluronic acid or hyaluronan, a ubiquitous carbohydrate polymer of host cells' extracellular matrix. Bacterial strains use this enzyme to break the polysaccharide that holds animal cells together and helps the host tissue spread. The primary function of hyaluronidase is to provide nutrients for the cell in gram-negative organisms. Recent studies have indicated that bacterial infection in the urinary tract region does not show hyaluronidase activity [174]. Some strains are known to cleave chondroitin sulfate [176]. Research work on enzyme hyaluronidase as a virulence factor in *Stenotrophomonas maltophilia* has not been done thoroughly.

3.4.2.4. Cytotoxins

Cytotoxicity is the production of toxic chemicals against cells. *S. maltophilia* is also known to produce cytotoxins. In a study conducted by Farrell *et al.*, cytotoxicity, primarily determined to be apoptotic cell death, was apparent in human lung epithelial cells treated with *S. maltophilia*. This demonstrates that *S. maltophilia* conveys cytotoxic factors and can instigate cytotoxicity in human lung epithelial cells. A study by Farrell *et al.* revealed that the cytotoxic effect of *S. maltophilia* on Vero, HeLa, and HEp-2 cells is different from any other toxin secreted by the bacteria [177]. The filtrates' action from culture supernatants of *S. maltophilia* results in loss of intercellular junctions, rounding, and blebbing, followed by the death of Hep 2 cell lines after 24 hours. The effect on Vero and HeLa cells was identified by initial rounding, rapid endocytosis, and cell aggregation within 30 min of incubation, with destruction and cell death after 72 hrs. This information concluded that Vero cells were more sensitive to cell lines, followed by HeLa cells. It was identified that the

cytotoxic activity is resistant to all protein inhibitors; however, it is thermolabile and can be inhibited by incubation at 56°C [177].

3.4.2.5. Hemolysin

Hemolysin production is significant when it comes to the virulence of glucose non-fermenting Gram-negative bacteria [130]. In a study conducted by Sanschagrin *et al.*, the hemolysis zones were determined in *S. maltophilia* after incubation at 37°C for 18-20 hours in aerobic and anaerobic incubators. A greenish hue resembling zones of α -hemolysis was detected in blood agar plates grown in both conditions [178]. In another study conducted by Sanschagrin *et al.*, it was reported that half of the strains tested had hemolytic activity. It was also demonstrated that lipase, protease, and lecithinase destroy cell membranes and correlate hemolytic with enzymatic activity [178]. Therefore, the results obtained above can be explained based on different erythrocyte cell membranes. When grown on different blood agar types, most *S. maltophilia* isolates presented hemolytic activity in rabbit blood more prominently after incubating for 96 hours at 37°C. β -hemolysis was observed more than α -hemolysis on different types of blood. It can be concluded that the hemolytic activity of *S. maltophilia* depends on blood type, incubation time, and temperature [179]. These findings concluded that *S. maltophilia* produces hemolysin according to the incubation period and substrate composition, resembling hot-cold hemolysins [180]. Hemolytic activity of β toxin is enhanced after incubation at temperatures below 10°C. Hence, this toxin is often referred to as the 'hot-cold hemolysin'.

3.4.2.6. Iron Acquisition Systems (Siderophores)

Iron is an essential compound that is required to perform various metabolic activities. It regulates oxidative stress response, biofilm formation, and different pathogenic mechanisms in *Stenotrophomonas maltophilia*. Iron is not soluble since it gets oxidized by environmental oxygen and becomes exceptionally insoluble. Bacteria have developed a low molecular weight, a ferric ion-specific chelating agent called siderophores, to acquire iron in its insoluble form [181]. These are small, high-affinity iron-chelating compounds with high specificity that bind to transferrin and lactoferrin proteins and serve as an iron transporter across the

cell membrane [182]. This is a prominent virulence factor for bacteria to survive in iron-restricted environments, as in hosts. In a study by Gordon *et al.*, siderophore production was screened by performing the chrome azurol S (CAS) agar assay. It has been observed that K279a and all clinical isolates were CAS-positive when grown on modified CAS agar plates for siderophore production. Arnov assay-catechol type and Csáky-Hydroxamate type were performed to determine the chemical nature of siderophores. It has been identified that the *S. maltophilia* strains associated with medical devices produce catechol-type siderophores; none of them is hydroxamate-type [183].

Further investigations on siderophores in *S. maltophilia* have proved that it possesses a distinct siderophore structure. It represents a novel catecholate, thereby notifying its unique iron acquisition mechanism [183]. On the other hand, Srijaruskul *et al.* concluded that *S. maltophilia* K279a produces the catechol-type compound enterobactin based on their recently sequenced genomes [184].

3.4.2.7. Nucleases

Production of RNase and DNase, the extracellular nucleases, has been demonstrated in many studies on the enzymatic activity of *S. maltophilia*. The ability to degrade DNA could be used for diagnosis, but the role of this ability has not yet been identified [185-187]. A study conducted in Brazil reported that after performing qualitative assays of *S. maltophilia* in nosocomial isolates, DNase activity was observed in 97.6% of tested strains. In a restricted carbohydrate environment, extracellular lipase converts lipids and participates in host tissue adhesion. Due to the lipase production by pathogens in lung infections, the virus is contributed in two ways: by hydrolyzing the lipid-rich tissue of the lungs or by inducing an intense inflammatory response [188]. After performing several studies on *S. maltophilia*, it has been concluded that it produces high amounts of lipases and contributes to pathogenesis [130, 137]. In CF patients, secretion of IgG antibodies due to lipase and phospholipase C has been observed [189]. The most toxic bacterial lecithinases display effects, such as platelet aggregation, including cytotoxic, myotoxic, and hemolytic activities. The degradation process occurs on two primary components of the outer leaflet of eukaryotic plasma membranes [190]. Ca^{2+} ions are very signifi-

cantly required for the interaction between phospholipases with phospholipid bilayers, and the sensitivity of erythrocytes from different species to lysis induced by phospholipases varies greatly. They hydrolyze phospholipids into fatty acids and destroy cell membranes, thereby contributing to the virulence of bacteria. The decreased hemolytic activity due to the presence of Zn^{2+} is similar to the results described by Berka and Vasil to PLC of *P. aeruginosa*. It has been demonstrated that all five culture supernatants of clinical isolates of *S. maltophilia* have activities similar to PLCs, and all five also present lecithinase activity. The entry of pathogens into cells and the disruption of normal cell function are both enabled by phospholipases. The role of phospholipases as a virulence factor is contributed by the destruction of mucus coating, lipoprotein membranes, and the host. Recent studies have found that urine isolates had no lecithinase activity and variable enzyme activity in isolates from tracheal fluid, sputum, cerebrospinal fluid, wound, and blood [190].

4. THERAPEUTIC APPROACH

S. maltophilia is resistant to many antibiotics, including tetracyclines, carbapenems, chloramphenicol, fluoroquinolones, TMP-SMX, and polymyxins. Low membrane permeability helps the bacteria resist β -lactams, such as ticarcillin-clavulanate, ceftazidime, and piperacillin-tazobactam. Ticarcillin-clavulanate has been proposed as an alternate therapy to TMP-SMX, but resistance to ticarcillin-clavulanate has been reported [191]. Studies reported that tigecycline and levofloxacin are new antimicrobial drugs that have proven effective in CF patients when used alone or in combination. Tigecycline works against the bacteria by inhibiting protein synthesis mechanisms [191]. Data suggested that co-trimoxazole should be the treatment of choice, despite the reports of the emergence of resistance but a combination treatment with co-trimoxazole, minocycline, and ticarcillin/clavulanate has also been suggested. In patients with bacteremia, intolerance, or allergy to TMP-SMX, ciprofloxacin combined with ticarcillin-clavulanate or ceftazidime can be used. New studies suggest that phage therapy, which lyses the bacterial cells, may be an alternative to antibiotics to treat *S. maltophilia* infections. The primary catechin Epigallocatechin-3-gallate (EGCG), found in green tea, has shown antimicrobial activity against *S.*

maltophilia. On studying 40 clinical isolates of *S. maltophilia*, it can be understood that the MICs range from 128 to 512 mg/L. The ability of *S. maltophilia* to adapt to its environment makes the pathogen a massive threat to the environment. Therefore, a different approach needs to be taken to deal with it. At the hospital and clinical level, it is essential to educate the medical staff. The risk of infection requires observation of aqueous-associated environments and regular cleaning and disinfection regimens for medical equipment surfaces that come into contact, directly or indirectly, with patients.

Antibiotic resistance and the virulence factors present in the bacteria restrict the use of antibiotics and therefore decline the therapeutic opportunities. Hence, a new treatment strategy that targets the immunogenic outer membrane protein of the bacteria has already been identified. The immunoproteomic approach can be used to identify the immunogenic outer membrane protein in *S. maltophilia*, which can be used effectively to treat and protect the *S. maltophilia* infection [192]. The experiment was carried out on mice, showing a significant result. Outer membrane proteins A (OmpA) and Smlt4123 were targeted. Mouse anti-Smlt4123 serum significantly reduced the bacterial counts in healthy individuals' blood in OPKA ($P < 0.05$), whereas the mouse anti-OmpA serum did not. The study revealed that the immunoproteomic approach has a better scope for treating *S. maltophilia* infections.

Epigallocatechin-3-gallate (EGCG) from green tea has been reported to interfere with *S. maltophilia* biofilm production as well as reduce its cell count *in vivo* [193]. A huge increase in dead cells within the biofilm produced by the bacteria in cystic fibrosis patients based on the EGCG dosage was observed under confocal laser scanning microscopy [193]. The studies showed significant results in both *in vivo* and *in vitro* techniques. This may be a novel therapeutic approach for drug resistance-related problems in *S. maltophilia* [194].

CONCLUSION

Despite advances in medical technologies and treatment methodologies, *S. maltophilia* infections require more understanding of the organism. More research and knowledge regarding alternative therapeutic agents in the management of *S. maltophil-*

ia infection are needed. *S. maltophilia* adapting to the local environment of the patient and altering the antimicrobial strategies to keep pace with the evolution of *S. maltophilia* are the most challenging aspects of antimicrobial drug designing. This requires the understanding of specific drug receptors.

CURRENT & FUTURE DEVELOPMENTS

The epidemiology and pathogenicity of *S. maltophilia* must be studied to a greater extent since this is the only way by which new drugs can be designed individually. Similarly, there is a need to evaluate the clinical setting that allows energy transfer in bacteria. Understanding nosocomial reservoirs and routes of transmission of the bacterium is required to develop effective infection control strategies. Shortly, it is to be noted that *S. maltophilia* will be a significant challenge to microbiologists, physicians, and hospital epidemiologists. For the prevention of *S. maltophilia* infection, many strategies have been proposed. Prevention is always better than cure-avoidance of inappropriate antibiotic use and prolonged external device implantation. Maintenance, proper disinfection, sterilization of ice-making machines, cardiopulmonary bypass apparatus, hemodialyzers, and respiratory therapy equipment are essential. Although carbonated water is acceptable, neutropenic patients should avoid drinking non-carbonated bottled water. During the infection of *S. maltophilia*, hand hygiene practices include washing hands with antimicrobial hand washes and wearing gloves when handling contaminated respiratory secretions.

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CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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